

HIGH LEVEL, AB, Canada

WMO: 710660

Lat: 58.620N

Lon: 117.160W

Elev: 1109

StdP: 14.12

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-31.4	-45.5	0.4	-37.2	-40.4	0.6	-31.1	19.1	7.5	17.9	4.5	0.5	0	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	81.9	63.0	78.8	61.8	75.8	60.4	65.6	77.4	63.8	75.3	62.1	72.8	6.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	84.0	70.4	59.3	78.4	68.4	57.5	73.6	66.7	31.0	77.5	29.7	75.6	28.4	73.1	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.3	13.5	DB	-42.9	87.6	6.2	2.9	-47.4	89.6	-51.1	91.3	-54.6	93.0	-59.1	95.1	(4)
(5)			WB	-42.6	68.9	5.6	2.2	-46.6	70.5	-49.9	71.8	-53.0	73.1	-57.0	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.3	-3.4	3.0	15.1	35.1	49.1	58.3	62.4	58.6	48.6	34.1	12.4	0.1		
	DBStd	25.96	15.50	13.43	15.06	10.73	8.48	6.01	4.92	5.95	7.43	8.31	12.32	14.18		
	HDD50	7918	1654	1316	1083	452	116	9	0	7	111	495	1127	1548		
	HDD65	12364	2119	1736	1548	898	494	210	108	210	493	958	1577	2013		
	CDD50	1082	0	0	0	5	88	259	385	274	68	3	0	0		
	CDD65	52	0	0	0	0	2	10	28	12	0	0	0	0		
	CDH74	1028	0	0	0	2	74	235	452	238	26	1	0	0		
	CDH80	175	0	0	0	0	10	34	89	41	1	0	0	0		
	WSAvg	5.1	4.3	4.8	6.1	6.7	6.4	5.5	4.8	4.5	5.0	5.1	4.4	3.9		
	PrecAvg	17.4	1.0	0.7	0.9	0.7	1.7	2.6	2.4	2.2	1.4	1.3	1.1	0.9		
(15) Precipitation	PrecMax	25.4	1.6	1.5	2.4	2.2	3.4	5.3	4.2	4.9	3.5	3.9	2.2	1.9		
	PrecMin	13.3	0.2	0.1	0.1	0.2	0.6	0.6	0.5	0.1	0.4	0.0	0.0	0.4		
	PrecStd	3.0	0.4	0.4	0.6	0.5	0.7	1.2	1.0	1.2	0.9	1.0	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	41.9	51.8	70.0	81.1	84.0	86.4	84.9	78.0	65.0	45.0	42.1		
		MCWB	33.9	34.7	40.3	50.5	58.3	62.7	65.8	65.2	61.1	51.5	37.2	36.0		
	2%	DB	31.5	34.6	46.0	62.3	75.4	79.8	82.4	79.8	72.1	56.6	37.6	32.3		
		MCWB	28.2	29.9	36.7	46.3	55.0	60.9	63.8	63.6	57.8	46.2	33.0	29.1		
	5%	DB	24.1	27.9	41.4	57.2	71.1	76.5	79.0	76.2	67.4	51.7	32.8	23.8		
		MCWB	22.6	24.7	34.1	43.0	52.9	59.3	62.9	62.0	54.9	43.0	30.7	22.5		
	10%	DB	16.9	22.1	36.8	52.9	66.8	73.1	75.7	72.8	63.4	47.4	29.1	19.0		
		MCWB	15.8	20.2	31.2	41.1	50.7	57.9	61.6	60.2	52.5	41.0	27.5	18.0		
	0.4%	WB	34.2	34.8	40.9	51.5	60.3	66.0	68.7	68.1	62.6	53.4	37.8	36.2		
		MCDB	39.7	41.0	50.6	68.9	76.2	78.0	79.6	80.2	75.8	63.2	43.4	42.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.3	29.9	37.0	46.9	56.7	63.0	65.9	65.2	59.0	46.6	33.6	29.0		
		MCDB	30.9	34.5	45.2	61.0	72.0	75.1	78.2	76.5	69.8	54.5	36.6	32.0		
	5%	WB	22.5	24.7	34.3	43.8	53.9	61.0	64.4	63.2	56.1	43.8	30.9	22.7		
		MCDB	24.2	27.8	41.1	55.5	68.3	73.0	76.2	73.9	65.0	50.5	33.1	24.0		
	10%	WB	15.8	20.2	31.3	41.4	51.6	59.2	62.8	61.2	53.6	41.2	27.5	17.9		
		MCDB	16.8	22.2	36.7	52.0	65.0	70.4	73.6	70.5	61.4	46.7	29.1	18.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	20.3	22.5	22.2	24.8	23.1	22.1	22.7	22.1	15.9	14.7	16.1		
		MCDBR	25.4	26.6	25.7	29.9	32.3	29.8	29.0	29.1	30.5	25.8	17.4	22.2		
		MCWBR	22.4	22.5	18.0	17.1	15.8	14.0	13.4	14.8	17.3	16.6	14.2	19.5		
		MCDBR	25.0	26.2	25.2	28.6	30.1	25.4	26.1	26.5	28.2	23.9	16.6	21.7		
	5% WB	MCWBR	22.1	22.4	18.0	16.9	15.1	12.7	13.0	14.3	16.7	15.9	13.7	19.2		
(40) Clear-Sky Solar Irradiance	taub		0.215	0.256	0.272	0.333	0.344	0.362	0.369	0.350	0.311	0.283	0.240	0.198		
	taud		2.241	2.276	2.290	2.314	2.423	2.401	2.398	2.443	2.554	2.561	2.284	2.112		
	Ebn at Noon		211	251	280	277	282	277	272	270	267	244	194	171		
	Edh at Noon		13	22	29	34	33	34	33	30	23	17	13	10		
(44) All-Sky Solar Radiation	RadAvg		175	469	1003	1550	1813	1902	1789	1449	979	473	185	105		
	RadStd		12	30	60	69	132	145	82	95	71	57	21	8		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+3.42	+5.02	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	+2.86	+3.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page